THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

December 16th, 2021 (Thursday) 2:30 PM – 5:30 PM (3 hours)

Answer only **5 (FIVE)** questions.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

Use the same answer book for both sections.
For the mechanics part, detailed reasoning, equations and steps are necessary.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

Q1 The compound shaft shown in Figure 1 is attached to rigid supports. For the bronze segment AB, the diameter is 75 mm, $\tau \leq 60$ MPa, and G = 35 GPa. For the steel segment BC, the diameter is 50 mm, $\tau \leq 80$ MPa, and G = 83 GPa.

(a) If a = 2 m and b = 1.5 m, compute the maximum torque T that can be applied. **(12 marks)**

(b) Determine the ratio of lengths a/b so that each material will be stressed to its permissible limit. What is the corresponding torque T? **(8 marks)**

Q2 A point on the surface of a machine is subjected to biaxial stress with $\sigma_x = 105\ MPa$, $\sigma_y = 75\ MPa$, and $\tau_{xy} = 25\ MPa$, as shown in Figure 3.

(a) Use plane stress transformation and equations to find the stresses acting on the plane oriented at an angle $\theta = 40^o$ from the x axia, where the angle θ is positive when counterclockwise. **(6 marks)**
(Hint:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} cos\,2\theta + \tau_{xy}\, sin\,2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} sin\,2\theta + \tau_{xy}\, cos\,2\theta \qquad)$$

(b) Use Mohr's Circle to find the stresses acting on the plane oriented at an angle $\theta = 40^o$ from the x axis, where the angle θ is positive when counterclockwise. **(8 marks)**

(c) Find the principal stresses and maximum shear stress, and their directions (showing stress state) using Mohr's Circle. **(6 marks)**

Q3 Two trusses, truss 1 and truss 2, are subject to point loads P_1 and P_2, as shown in Figure 3 (a) and (b) respectively.

(a) Calculate the axial forces of all the bars in truss 1 and truss 2. The axial forces should be expressed in terms of P_1 or P_2. **(10 marks)**

(b) Use Euler's buckling theory to calculate the critical buckling loads $P_{1,cr}$ and $P_{2,cr}$ of the two trusses, respectively. Also point out which bars will buckle for the two trusses. For each of the members, EI = 300 kN m^2. **(10 marks)**

FIGURES

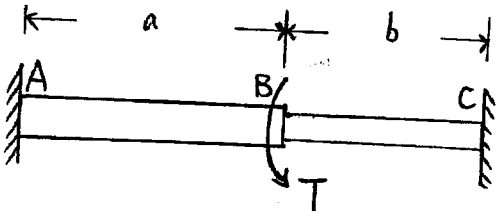


Figure 1

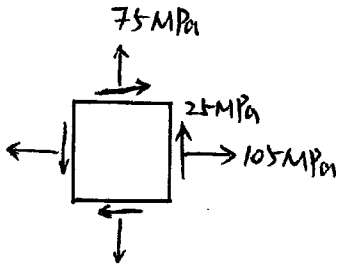


Figure 2

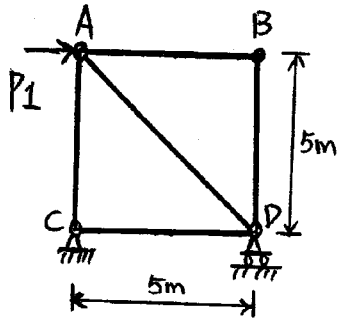


Figure 3(a) Truss 1

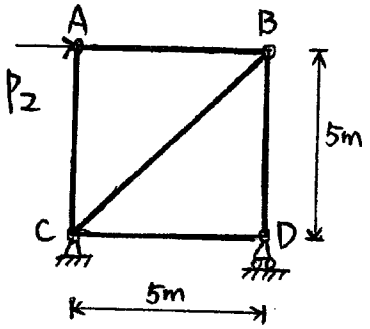


Figure 3(b) Truss 2

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